

7(a) (i)	electron	B1
7(a) (ii)	any two: • can be deflected by electric and magnetic fields or negatively charged • absorbed by few (1 – 4) mm of aluminum / 0.5 to 2 m or metres for range in air • speed up to 0.99c / range of speeds / energies	B2
7(a) (iii)	decay occurs and cannot be affected by external / environmental factors such as chemical / pressure / temperature / humidity (any two stated factors)	B1
7(b)	3 and 0 for superscript numbers 2 and –1 for subscript numbers	B1
7(c)	energy = $5.7 \times 10^3 \times 1.6 \times 10^{-19}$ (= 9.12×10^{-16} J) $v^2 = \frac{2 \times 9.12 \times 10^{-16}}{9.11 \times 10^{-31}}$ $v = 4.5 \times 10^7 \text{ m s}^{-1}$	C1 A1
8(a)	It represents the rated power output that the turbine is expected to generate	B1